Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/06/2025
PI Name:	Patel, Zarana Ph.D.		
Project Title:	Development of an Autonomous, Dual Chamber Bioreactor for the Growth of 3-Dimensional Epithelial-Stromal Tissues in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	zarana.s.patel@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-3723
Organization Name:	KBRwyle/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	NOTE: PI moved to Wyle in 2014; previously at Universities Space Research Association.		
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/05/2014	**End Date:**	09/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huff, Janice (Universities Space Research Association)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE (9/10/2014): PI has changed institutions and work continues as an internal NASA Johnson Space Center project. Previous title was "Development of an Autonomous, Dual Chamber Bioreactor for the Growth of 3-Dimensional Epithelial-Stromal Tissues in Microgravity." We propose the development of a novel, autonomous bioreactor that can provide for the growth and maintenance in microgravity of 3-D organotypic epithelial-stromal cultures that require an air-liquid interface. These complex 3-D tissue models accurately represent the morphological features, differentiation markers, and growth characteristics observed in normal human epithelial tissues, including the skin, esophagus, lung, breast, pancreas, colon. However, because of their precise and complex culture requirements, such as an air-liquid interface, these 3-D models have yet to be utilized for life sciences research aboard the International Space Station. The proposed technology will provide the capability to perform biological research on the ISS using these realistic, tissue-like human epithelial-stromal cell models and will contribute significantly to advances in fundamental space biology research on questions regarding microgravity effects on normal tissue development, aging, cancer and other disease processes. It will also allow for the study of how combined stressors, such as microgravity with radiation and nutritional deficiencies, affect multiple biological processes and provides a platform for conducting countermeasure investigations on the ISS without the use of animal models. The final design will be autonomous and consist of a dual chamber that provides for air-liquid, liquid-liquid, and liquid-air exchanges within the cell culture units. The bioreactor will support multiple tissue types and its modular design will provide for incorporation of add-on capabilities such as microfluidics drug delivery, media sampling, and in situ biomarker analysis. Finally, this proposal leverages the design of an existing 2nd generation prototype which has already been tested on parabolic flights through NASA's Flight Opportunities Program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology will provide the capability to perform biological research on the ISS using a realistic, tissue-like human epithelial-stromal cell model that accurately represents the morphological features, differentiation markers, and growth characteristics observed in normal human epithelial tissues. The enabling technology will contribute significantly to advances in fundamental space biology research on questions regarding microgravity effects on normal tissue development, aging, cancer and other disease processes.
Task Progress:	Temp
Bibliography Type:	Description: (Last Updated: 08/25/2020)